

XUAN LIU

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EDUCATION

Washington University in St. Louis

St. Louis, Missouri, USA

Ph.D. Energy, Environmental & Chemical Engineering

2019 – 2024

• Supervisors: Prof. Randall Martin, Prof. Jay Turner

• GPA: 3.96/4.00

Xiamen University

Xiamen, Fujian, China

B.S. Chemical Engineering and Technology

2015 – 2019

• Overall GPA: 3.86/4.00; Major GPA: 3.94/4.00

• Rank: 1/60

RESEARCH EXPERIENCE

Postdoctoral Scholar at University of California San Diego, Advisors: Amato Evan, William Porter

- Assess the impacts of dust from ephemeral lakes on regional weather and climate using the newly developed soil mineralogy dataset from NASA's Earth surface Mineral dust source InvesTigation (EMIT) mission (2024 – present)
- Examine changes in exposure to playa dust as the Salton Sea continues to shrink using the Weather Research and Forecasting Model with Chemistry (WRF-Chem) (2025 – present)
- Improve WRF-Chem simulations of dust in the Coachella Valley by constraining emissions with Portable In-Situ Wind ERosion Laboratory (PI-SWERL) measurements to identify key sources and inform mitigation efforts (2026 – present)

Graduate Research Assistant at Washington University in St. Louis, Advisors: Randall Martin, Jay Turner

- Investigated the characteristics of airborne dust and trace elements in Uzbekistan with measurements from filter samples, low-cost sensors, and soil samples (2022 – 2024)
- Examined the attenuation effect due to particle size and mass loading for crustal elements in X-ray Fluorescence analysis of ambient particulate matter samples (2022 – 2024)
- Developed the methodology and explored implications for elemental analysis of ambient particulate matter samples from the global Surface PARTiculate mATter Network (SPARTAN) (2021 – 2023)
- Developed and evaluated a global-scale mineral dust equation with region-specific coefficients for estimating dust mass (2020 – 2022)

Undergraduate Research Assistant at Xiamen University, Advisors: Qiugen Zhang, Gaofeng Zheng, Song Tu

- Developed a new type of side-chain-type polymer anion exchange membranes for fuel cells (2017 – 2018)
- Applied electrospinning to prepare polymer nanofibers for heavy metal ion adsorption (2016 – 2018)
- Investigated optimal purification conditions of 2-imidazolidinone (2016 – 2018)

PUBLICATIONS

- **Liu, X.**, Ginoux, P., Pérez García-Pando, C., González-Romero, A., Querol, X., Brodrick, P. G., & Evan, A. (2026). Comparison of soil mineralogy from EMIT imaging spectroscopy and ground-based measurements in the deserts of southern California. *Journal of Geophysical Research: Atmospheres*, 131(12), e2026JD046595. <https://doi.org/10.1029/2026JD046595>
- **Liu, X.**, Turner, J. R., Mitroo, D., Ren, Y., Oxford, C. R., Liu, W., & Martin, R. V. (2025). Assessing Attenuation Effects in X-ray Fluorescence Analysis of Light Elements in Mineral Dust. *ACS ES&T Air*, 3(1), 175-185. <https://doi.org/10.1021/acsestair.5c00295>
- **Liu, X.**, Amonov, M., & Turner, J. R. (2025). Quantifying Dust Contribution to Particulate Matter in Central Asia: Insights from the Elemental Composition of PM_{2.5} and PM₁₀ in Uzbekistan. *Atmospheric Pollution Research*, <https://doi.org/10.1016/j.apr.2025.102781>
- **Liu, X.**, Turner, J. R., Oxford, C., McNeill, J., Walsh, B., Roy, E. L., ... Martin, R. V. (2024). Elemental Characterization of Ambient Particulate Matter for a Globally Distributed Monitoring Network: Methodology and Implications. *ACS EST Air*, 1(4), 283-293. <https://doi.org/10.1021/acsestair.3c00069>

- **Liu, X.,** Turner, J. R., Hand, J. L., Schichtel, B. A., Martin, R. V. (2022). A Global-Scale Mineral Dust Equation. *Journal of Geophysical Research: Atmospheres*, 127(18), e2022JD036937. <https://doi.org/10.1029/2022JD036937>
- Eom, S., Yeo, M. J., Lee, D., Koo, J. H., Kim, J., Ren, Y., Oxford, C. R., **Liu, X.,** ... & Park, S. S. (2026). Contrasting long-range transport signals of metropolitan PM_{2.5} in South Korea detected by SPARTAN observations. *Atmospheric Pollution Research*, 17(8), 103088. <https://doi.org/10.1016/j.apr.2026.103088>
- Zhang, D., Martin, R. V., **Liu, X.,** Van Donkelaar, A., Oxford, C. R., Li, Y., ... & Windwer, E. (2025). Improving annual fine mineral dust representation from the surface to the column in GEOS-Chem 14.4. 1. Geoscientific Model Development, 18(19), 6767-6803. <https://doi.org/10.5194/gmd-18-6767-2025>
- Eom, S., Park, S. S., **Liu, X.,** Oxford, C. R., Martin, R. V., Kim, J., & Song, C. K. (2025). Impact of Chemical Composition on Aerosol Scattering: Insights from the Surface Particulate Matter Network and Aerosol Robotic Network. *Environmental Science & Technology*, 59(37), 19899-19909. <https://doi.org/10.1021/acs.est.5c09325>
- Ren, Y., Oxford, C. R., Zhang, D., **Liu, X.,** Zhu, H., Dillner, A. M., ... & Martin, R. V. (2025). Black carbon emissions generally underestimated in the global south as revealed by globally distributed measurements. *Nature communications*, 16(1), 7010. <https://doi.org/10.1038/s41467-025-62468-5>
- Li, C., Martin, R. V., van Donkelaar, A., Jimenez, J. L., Zhang, Q., Turner, J. R., **Liu, X.,** ... & Thurston, G. D. (2025). Estimates of submicron particulate matter (PM₁) concentrations for 1998–2022 across the contiguous USA: leveraging measurements of PM₁ with nationwide PM_{2.5} component data. *The Lancet Planetary Health*, 9(6), e491-e502. [https://doi.org/10.1016/S2542-5196\(25\)00094-4](https://doi.org/10.1016/S2542-5196(25)00094-4)
- Zhu, H., Martin, R. V., Van Donkelaar, A., Hammer, M. S., Li, C., Meng, J., Oxford, C. R., **Liu, X.,** ... & Lyapustin, A. (2024). Importance of aerosol composition and aerosol vertical profiles in global spatial variation in the relationship between PM_{2.5} and aerosol optical depth. *Atmospheric Chemistry and Physics*, 24(20), 11565-11584. <https://doi.org/10.5194/acp-24-11565-2024>
- Perez-Plazola, M., Diaz, J., Stewart, C. P., Arnold, C. D., Caswell, B. L., Lutter, C. K., Werner, E. R., Maleta, K., Turner, J., Prathibha, P., **Liu, X.,** Gyimah, E., Iannotti, L. (2023). Plasma Mineral Status after a Six-Month Intervention Providing One Egg per Day to Young Malawian Children: A Randomized Controlled Trial. *Scientific Reports*, 13(1), 6698. <https://doi.org/10.1038/s41598-023-33114-1>
- Lin, C., **Liu, X.,** Yang, Q., Wu, H., Liu, F., Zhang, Q., Zhu, A., Liu, Q. (2019). Hydrophobic Side Chains to Enhance Hydroxide Conductivity and Physicochemical Stabilities of Side-Chain-Type Polymer AEMs. *Journal of Membrane Science*, 585, 90–98. <https://doi.org/10.1016/j.memsci.2019.04.066>

PRESENTATIONS

- EMIT Science and Applications Technical Interchange Meeting (2026), *Using EMIT to Improve the Representation of Dust and Assess Its Impacts with WRF-Chem*. Pasadena, CA. **Talk.**
- American Geophysical Union Fall Meeting (AGU 2025), *Comparison of soil mineralogy from EMIT imaging spectroscopy and ground-based measurements in the deserts of southern California*. New Orleans, LA. **Talk.**
- EMIT Science and Applications Technical Interchange Meeting (2025), *Comparison of Soil Mineralogy from EMIT Remote Imaging Spectroscopy and Ground-Based X-ray Diffraction Measurements in Southern California Deserts*. Pasadena, CA. **Talk.**
- American Geophysical Union Fall Meeting (AGU 2024), *Assessing Attenuation Effects in X-ray Fluorescence Analysis of Light Elements in Mineral Dust for Ground-Based Measurements*. Washington, D.C. **Poster.**
- American Geophysical Union Fall Meeting (AGU 2023), *Elemental Composition of Fine Particulate Matter in Central Asia: Dust and Trace Elements in Tashkent, Uzbekistan*. San Francisco, CA. **Poster.**
- American Association for Aerosol Research Annual Conference (AAAR 2023), *Elemental Characterization of Ambient Particulate Matter for a Globally Distributed Monitoring Network: Methodology and Implications*. Portland, OR. **Talk.**
- The 4th International Surface PARTICulate mAtter Network (SPARTAN) Meeting (2023), *Advances in elemental characterization with the XRF*. St Louis, MO. **Talk.**
- American Geophysical Union Fall Meeting (AGU 2022), *Elemental Characterization of Ambient Particulate Matter for a Globally Distributed Monitoring Network: Methodology and Implications*. Chicago, IL. **Talk.**
- American Association for Aerosol Research Annual Conference (AAAR 2021), *A Global-scale Mineral Dust Equation*. Virtual. **Talk.**

- Atmospheric Optics: Aerosols, Visibility, and the Radiative Balance Conference (**2021**), *A Global-scale Mineral Dust Equation*. Bryce Canyon, UT. **Talk**.
- American Geophysical Union Fall Meeting (**AGU 2020**), *A Global-scale Mineral Dust Equation*. Virtual. **Poster**.

SELECTED AWARDS & HONORS

- Second Prize in the 26th annual Graduate Research Symposium, Washington University in St. Louis Graduate Student Senate and Graduate Professional Council, 2021
- Outstanding Graduate, Xiamen University, 2019
- National Scholarship, Ministry of Education of P. R. China, 2017 & 2018
- National First Prize in the 12th National Undergraduate Chemical Design Competition, National Undergraduate Chemical Design Competition Committee, 2018
- Merit Student, Xiamen University, 2016 – 2019

TEACHING & ADVISING EXPERIENCE

Teaching Assistant, Washington University in St. Louis

Air Quality Engineering with Lab (Fall 2022); Kinetics and Reaction Engineering Principles (Spring 2021); Unit Operations Laboratory (Fall 2020)

Student Mentor, Washington University in St. Louis

Mentored graduate (Yingxi Lu, Xinying Wang, Yuxuan Ren) and undergraduate (Caroline DeLuca, Yuanlong Su, Ryan Graham) students in air quality research, including training in instrumentation, laboratory experiments, data analysis, and interpretation

Student Mentor, University of California San Diego

Mentored graduate student (Amanda Mallinckrodt) in dust mineralogy research

PROFESSIONAL ACTIVITIES & SERVICE

- **Committee Member**, University of California Dust Symposium, 2026
- **Session Chair**, AAAR Annual Conference, 2023
- **Site Manager**, SPARTAN Network, 2022–2024 — Coordinated six international monitoring sites in Abu Dhabi, UAE; Dhaka, Bangladesh; Mexico City, Mexico; Seoul, South Korea; Sherbrooke, Canada; and Ulsan, South Korea, including operator communication, sample shipments, data management, and troubleshooting
- **Journal Reviewer**, Journal of Geophysical Research: Atmospheres, Remote Sensing of Environment, GeoHealth, Atmospheric Chemistry and Physics, ACS Chemical Health & Safety, Aerosol Science & Technology

OUTREACH

- San Diego County Office of Education STEM Design Team Summer Academy — Engaged K–12 educators and Student Fellows through a research presentation and hands-on field-study demonstration, introducing participants to scientific instrumentation and environmental research methods, 2026
- WashU Center for Aerosol Science and Engineering (CASE) Student Chapter Committee — Organized events and activities to foster student engagement, professional development, and knowledge exchange in aerosol science, 2023–2024

TECHNICAL SKILLS

- Field and analytical methods: Portable In-Situ Wind EROsion Lab (PI-SWERL), ambient particulate matter sampling, X-ray Fluorescence (XRF), Inductively Coupled Plasma Mass Spectrometry (ICP-MS)
- Atmospheric modeling: Weather Research and Forecasting Model with Chemistry (WRF-Chem)
- Programming and data analysis: R, Python, MATLAB
- Geospatial and remote sensing: ArcGIS, ENVI